

DEVELOPING APPS ON QT



In the last article, we worked on the backbone of GUIs, the signal and slot mechanism. In this article, we move ahead to the GUI part of Qt.

Part—4

The basic block of any GUI application is the widget, which provides the functionality of GUI components like buttons, lists, etc. In Qt, *QWidget* is the base class of all other GUI classes. It provides the main functionality, like keyboard and mouse events.

One of the simplest, yet most important widgets is the button. For example, in Figure 1, when I tried to close Firefox with multiple tabs opened, I was prompted by a dialogue box with two push-buttons, *Cancel* and *Close Tabs*. Without buttons, a GUI would be useless. All button widgets are derived from the *QAbstractButton* class, which provides four signals—*clicked()*, *pressed()*, *released()* and *toggled()*, and provides five states—*isChecked()*, *isEnabled()*, *isDown()*, *setCheckable()* and *setAutoRepeat()*. We will cover these in the sample code. Some of the classes derived from *QAbstractButton* are: *QPushButton*, *QRadioButton* and *QCheckBox*.

Let's start with *QPushButton*. This widget is used to let the user initiate some action, like *OK*, *Cancel*, *Apply*, *Save*, etc. The *QPushButton* widget emits *clicked()*, *pressed()* and *released()* signals. We will use the signals and slot mechanism covered in the last article, to demonstrate the use of buttons.

Create a GUI application and drag-drop a *PushButton* on the dialogue window. You can change the name from the default *pushButton* by right-clicking the widget, then select *Change objectName*, and give it a new name. Similarly, you can rename the dialogue window and other classes.

Download and experiment on it.

Here is sample code for the *QPushButton* class (*dialog.h*):

```
#ifndef DIALOG_H
#define DIALOG_H
#include <QDialog>
namespace Ui {
    class Dialog;
}
class Dialog: public QDialog
{
    Q_OBJECT
public:
    explicit Dialog(QWidget *parent = 0);
    ~Dialog();
private:
    Ui::Dialog *ui;
```



Note: The sample code from this article is available at http://www.linuxforu.com/article_source_code/apr12/qt_4_source_code.zip

```
private slots:
    void button_clicked();
    void button_pressed();
    void button_released();
};
```